LAPLACE TRANSFORMATION — ENGINEERING MATHEMATICS CHEATSHEET

Core formulas • Transform methods • Inverse transforms • Initial-value problem workflow

1. Definition and Fundamental Transform Pairs

Function f(t), t ≥ 0	Laplace transform F(s) = ℒ{f(t)}	Conditions / note
1	$1/s$	Re(s) > 0
t^n (n = 0,1,2,...)	$n! / s^{n+1}$	Re(s) > 0
e^{at}	$1/(s - a)$	Re(s) > a
sin(at)	$a/(s^2 + a^2)$	
cos(at)	$s/(s^2 + a^2)$	
sinh(at)	$a/(s^2 - a^2)$	
cosh(at)	$s/(s^2 - a^2)$	
e^{at} sin(bt)	$b/((s - a)^2 + b^2)$	Frequency shift
e^{at} cos(bt)	$(s - a)/((s - a)^2 + b^2)$	Frequency shift
δ(t − a), a ≥ 0	e^{-as}	Impulse
u(t − a)	e^{-as}/s	Unit step

2. Properties and Operational Rules

Property	Formula	Use
Linearity	$\mathcal{L}\{af + bg\} = aF + bG$	Split / combine terms
First derivative	$\mathcal{L}\{f'\} = sF(s) - f(0^+)$	Initial value
Second derivative	$\mathcal{L}\{f''\} = s^2F - sf(0^+) - f'(0^+)$	ODEs
nth derivative	$\mathcal{L}\{f^{(n)}\} = s^nF - \Sigma_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^+)$	General ODE
Multiplication by t	$\mathcal{L}\{t f(t)\} = -dF/ds$	Repeated: $\mathcal{L}\{t^n f\}=(-1)^n F^{(n)}$
Integration	$\mathcal{L}\{\int_0^t f(\tau)d\tau\} = F(s)/s$	Accumulation
First shifting	$\mathcal{L}\{e^{at} f(t)\} = F(s - a)$	Shift $s \longrightarrow s - a$
Second shifting	$\mathcal{L}\{u(t - a)f(t - a)\} = e^{-as}F(s)$	Delayed signal
Convolution	$\mathcal{L}\{f * g\} = F(s)G(s)$	$f*g = \int_0^t f(\tau)g(t-\tau)d\tau$
Periodic f, period T	$\mathcal{L}\{f\} = [\int_0^T e^{-st}f(t)dt]/[1 - e^{-sT}]$	One period only

3. Initial- and Final-Value Theorems

- Initial value: $f(0^+) = \lim_{s\to\infty} sF(s)$, when the limit exists.
- Final value: $f(\infty) = \lim_{s\to 0} sF(s)$, provided all poles of sF(s) lie in the open left half-plane (no RHP poles or nonzero imaginary-axis poles).
- Use the theorems as checks—not as substitutes for finding f(t) when the full response is required.

LAPLACE TRANSFORMATION — METHODS & SOLUTION PROCEDURES

Choose the method from the algebraic structure of F(s) or the differential equation

4. Inverse Laplace Transform: Method Selection

Form / situation	Method	Procedure
Rational F(s) with distinct linear factors	Partial fractions	Factor denominator → decompose → use $1/(s-a) \leftrightarrow e^{at}$.
Repeated pole $(s-a)^m$	Repeated partial fractions	Write $A_1/(s-a)+\ldots+A_m/(s-a)^m$ → solve coefficients → use $t^n e^{at}/n!$ pair.
Irreducible quadratic	Complete square	Rewrite $s^2+2as+(a^2+b^2)$ → match cosine / sine pairs.
Shifted quadratic / numerator	Algebra + frequency shift	Express numerator using (s−a) and constants; apply e^{at} shift.
Product F(s)G(s), not readily decomposed	Convolution	Find f,g → evaluate $\int_0^t f(\tau)g(t-\tau)d\tau$.
$e^{-as}F(s)$	Second shifting	Invert F(s) to f(t); answer u(t−a)f(t−a).
Piecewise / delayed input	Unit-step representation	Rewrite input with u(t−a) and shifted function; transform each term.

5. Partial-Fraction Decomposition: Template

Denominator factor	Required expansion
Distinct linear: (s−a)(s−b)	$A/(s-a) + B/(s-b)$
Repeated linear: $(s-a)^m$	$A_1/(s-a) + A_2/(s-a)^2 + \ldots + A_m/(s-a)^m$
Irreducible quadratic: s^2+ps+q	$(As+B)/(s^2+ps+q)$
Repeated irreducible quadratic	$(A_1s+B_1)/Q + (A_2s+B_2)/Q^2 + \ldots$

- Cover-up (simple distinct pole): $A = \lim_{s\to a} (s-a)F(s)$.
- Repeated poles: equate coefficients after clearing denominators, or differentiate the appropriate expression.
- Then map terms: $1/(s-a)^n \leftrightarrow t^{n-1}e^{at}/(n-1)!$.

6. Solving Linear ODE Initial-Value Problems — Standard Workflow

Step	Procedure
1. Identify	Write the ODE and initial data y(0), y′(0), … clearly.
2. Transform	Apply $\mathcal{L}$ to both sides; substitute derivative formulas using initial values.
3. Solve algebraically	Collect Y(s); isolate Y(s) = numerator / denominator.
4. Simplify	Factor denominator; use partial fractions, completing square, or shifts.
5. Invert	Apply the transform table and properties term-by-term.
6. Verify	Check initial conditions; optionally use initial/final-value theorem where valid.

7. Worked Procedure (illustrative): $y' + 2y = 1$, $y(0)=0$

- Transform: $sY(s) - 0 + 2Y(s) = 1/s$.
- Isolate: $Y(s) = 1/[s(s+2)]$.
- Partial fractions: $Y(s) = (1/2)(1/s) - (1/2)(1/(s+2))$.
- Inverse: $y(t) = 1/2 - 1/2\, e^{-2t} = (1 - e^{-2t})/2$, $t \geq 0$.
- Check: $y(0)=0$; as $t\to\infty$, $y\to 1/2$.

8. Common Errors — Quick Check

- Do not omit initial-condition terms in $\mathcal{L}\{y'\}$, $\mathcal{L}\{y''\}$, etc.
- Second shifting requires the shifted function f(t−a) multiplied by u(t−a); it is not simply $e^{-as}F(s)$ for an unshifted f(t).
- Complete the square carefully; match the numerator to s−a or a constant before using sine/cosine pairs.
- Check factorization, signs, repeated-pole powers, and units / domain $t \geq 0$.
- Final-value theorem is invalid for unstable responses or poles on the imaginary axis (except carefully treated special cases).